MT23CE04A:: Cloud Computing

Course Type	MC	Semester	6

Teaching Scheme		Credits		Examination Scheme	
Lecture	3Hr./Week	Lecture	3	CIE Marks	60
Tutorial	--Hr./Week	Tutorial	-	ESE Marks	40
Practical/Studio	0Hr./Week	Practical/Studio	0	Total Marks	100
Total	3Hr./Week	Total	3		

Course Description

This course gives a deep insight into cloud computing principles, models, and services. It includes cloud architecture, virtualization, security, and hands-on practice with major cloud platforms like AWS, Azure, and Google Cloud. Students will learn about cloud storage, security issues, and emerging trends in cloud technology, which include edge computing and serverless architectures.

Course Outcomes

CO No.	Statement
1	Explain the fundamental concepts, characteristics, and benefits of cloud computing.
2	Understanding cloud architecture and security aspects.
3	Demonstrate knowledge of virtualization technologies and their role in cloud computing.
4	Synthesis of different cloud simulators for real time application development.
5	Deployment of applications on cloud platforms .

Mapping of COs to POs and PSOs

CO No			Pos										PSOs				BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4	
CO1	3																UN
CO2		2															UN
CO3					2												AP
CO4									2								EV
CO5															3		CR

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No 1	Introduction to Cloud Computing	Hours 9	CO	BTL
Basics of Distributed and Parallel Computing , Basics of Distributed and Parallel Computing ,Evolution of Cloud Computing, Characteristics of Cloud Computing, Benefits and Limitations of Cloud Computing, Cloud Computing vs. Traditional Computing, Cloud Adoption Challenges, Applications of Cloud Computing			1	UN
Unit No 2	**Cloud Architecture and Security**	**Hours 9**	**2**	**AN**
Cloud Computing Architecture ,Virtualization and Its Role in Cloud Computing ,Cloud Service Models: Infrastructure as a Service (IaaS) ,Platform as a Service (PaaS) ,Software as a Service (SaaS) ,Cloud Deployment Models: Public Cloud ,Private Cloud ,Hybrid Cloud ,Community Cloud . Cloud Security Models ,Identity and Access Management (IAM) ,Threats in Cloud Security and Mitigation Strategies.				
Unit No 3	**Virtualization and Resource Management**	**Hours 9**	**3**	**UN**
Basics of Virtualization , Types of Virtualizations: Compute, Storage, and Network Hypervisors: Type 1 & Type 2 ,Virtual Machines vs. Containers , Docker and Kubernetes Overview, Resource Allocation and Load Balancing.				
Unit No 4	Cloud Simulators	**Hours 9**	**3**	**UN**
CloudSim and GreenCloud : Introduction to Simulator, understanding CloudSim simulator, CloudSim Architecture(User code, CloudSim, GridSim, SimJava) Understanding Working platform for CloudSim, Introduction to GreenCloud.				
Unit No 5	**Cloud Platforms, Applications, and Emerging Trends**	**Hours 9**	**5**	**AP**
Overview of Cloud Service Providers: AWS, Azure, Google Cloud ,Hands-on with Cloud Services: Compute (EC2, Virtual Machines, Lambda) ,Storage (S3, Blob Storage, Google Cloud Storage) , Databases (DynamoDB, Cloud SQL) Cloud in IoT, AI, and Big Data ,Serverless Computing & Edge Computing Cloud-Native Development and microservices.				

Textbooks

1	*Rajkumar Buyya, Christian Vecchiola, S. Thamarai Selvi, "Mastering Cloud Computing", McGraw Hill Education, 2013.*
2	*Kai Hwang, Geoffrey C. Fox, Jack J. Dongarra, "Distributed and Cloud Computing", Morgan Kaufmann, 2011.*
3	*"Cloud Computing: A Hands-On Approach" Arshdeep Bahga, Vijay Madisetti 2023*

Reference Books / Journal Articles / Weblink

1	*"Cloud Computing: Concepts, Technology & Architecture" Thomas Erl, Zaigham Mahmood, and Ricardo Puttini 2013 Pearson Education*
2	*"Architecting the Cloud: Design Decisions for Cloud Computing Service Models (SaaS, PaaS, and IaaS)" – Michael J. Kavis (2023)*
3	*"Cloud Computing" Nayan B. Ruparelia 2023 The MIT Press*